MPG No. 6.4.123 2′2′2 (22'2' setting) [Type III, orthorhombic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} M_x & 0 & 0 \end{bmatrix}$$

$$M_x = \mathbb{M}_x^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{yz} \\ 0 & M_{yz} & 0 \end{bmatrix}$$

$$M_{yz} = \mathbb{M}_{yz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{yz} \\ 0 & -M_{yz} & 0 \end{bmatrix}$$

$$M_{yz} = \mathbb{T}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & 0 \\ M_{yyx} & 0 & 0 \\ M_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{zzx} \\ 0 & M_{yyx} & 0 \end{bmatrix}$$

$$M_{xxx} = 3\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)}$$
$$M_{yyx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\beta(1)}$$
$$M_{zzx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\beta(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ M_{yyx} & 0 & 0 \\ M_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{M_{zzx}}{2} \\ 0 & -\frac{M_{yyx}}{2} & 0 \end{bmatrix}$$

$$M_{yyx} = 2\mathbb{M}_x^{(2)} + 2\mathbb{T}_{yz}^{(1)}$$
$$M_{zzx} = 2\mathbb{M}_x^{(2)} - 2\mathbb{T}_{yz}^{(1)}$$
$$\frac{M_{zzx}}{2} = \mathbb{M}_x^{(2)} - \mathbb{T}_{yz}^{(1)}$$
$$\frac{M_{yyx}}{2} = \mathbb{M}_x^{(2)} + \mathbb{T}_{yz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{zxz} \\ 0 & M_{xyy} & 0 \end{bmatrix}$$

$$M_{zxz} = \mathbb{M}_{x}^{(3)} + \mathbb{T}_{yz}^{(2)}$$
$$M_{xyy} = -\mathbb{M}_{x}^{(3)} + \mathbb{T}_{yz}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & M_{yyyz} & 0 & 0 \\ 0 & 0 & 0 & M_{zzyz} & 0 & 0 \\ M_{xxyz} & M_{yyyz} & M_{zzyz} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{zxxy} \\ 0 & 0 & 0 & 0 & M_{zxxy} & 0 \end{bmatrix}$$

$$M_{xxyz} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)}$$
$$M_{yyyz} = \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)}$$
$$M_{zzyz} = -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)}$$
$$M_{zxxy} = 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & M_{yyyz} & 0 & 0 \\ 0 & 0 & 0 & M_{zzyz} & 0 & 0 \\ -M_{xxyz} & -M_{yyyz} & -M_{zzyz} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{zxxy} \\ 0 & 0 & 0 & 0 & -M_{zxxy} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{M}_{yz}^{(3)} - 4\mathbb{T}_{x}^{\beta(1)}$$
$$M_{yyyz} = \mathbb{M}_{yz}^{(3)} + 2\mathbb{T}_{x}^{(1)} - 2\mathbb{T}_{x}^{\alpha(1)} + 2\mathbb{T}_{x}^{\beta(1)}$$
$$M_{zzyz} = \mathbb{M}_{yz}^{(3)} - 2\mathbb{T}_{x}^{(1)} + 2\mathbb{T}_{x}^{\alpha(1)} + 2\mathbb{T}_{x}^{\beta(1)}$$
$$M_{zxxy} = -\mathbb{T}_{x}^{(1)} - 4\mathbb{T}_{x}^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{zxxy} \\ 0 & M_{zxxy} & 0 \end{bmatrix}$$

$$M_{zxxy} = 2\mathbb{M}_{yz}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{zxxy} \\ 0 & -M_{zxxy} & 0 \end{bmatrix}$$

$$M_{zxxy} = -\mathbb{T}_{x}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xxyz} & 0 & 0 \\ M_{yyyz} & 0 & 0 \\ M_{zzyz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{zxxy} \\ 0 & M_{xyzx} & 0 \end{bmatrix}$$

$$
\begin{aligned}
M_{xxyz} &= \mathbb{T}_x^{(3)} + 2\mathbb{T}_x^{(4)} + 2\mathbb{T}_x^{\alpha(2)} \\
M_{yyyz} &= -2\mathbb{M}_{yz}^{(5)} + \mathbb{T}_x^{(3)} - \mathbb{T}_x^{\alpha(2)} + \mathbb{T}_x^{\beta(2)} \\
M_{zzyz} &= 2\mathbb{M}_{yz}^{(5)} + \mathbb{T}_x^{(3)} - \mathbb{T}_x^{\alpha(2)} - \mathbb{T}_x^{\beta(2)} \\
M_{zxxy} &= -\mathbb{M}_{yz}^{(5)} + \mathbb{T}_x^{(4)} - \mathbb{T}_x^{\alpha(2)} - \mathbb{T}_x^{\beta(2)} \\
M_{xyzx} &= \mathbb{M}_{yz}^{(5)} + \mathbb{T}_x^{(4)} - \mathbb{T}_x^{\alpha(2)} + \mathbb{T}_x^{\beta(2)}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & M_{zxxy} \\
0 & 0 & 0 & 0 & M_{xyzx} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{yzxx} &= \mathbb{T}_x^{(5)} + 2\mathbb{T}_x^{(6)} + 2\mathbb{T}_x^{\alpha(3)} \\
M_{yzyy} &= -2\mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(5)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{yzzz} &= 2\mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(5)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)} \\
M_{zxxy} &= \mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{xyzx} &= -\mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)}
\end{aligned}
$$

* Rank 4 tensor (s).

$$
\begin{bmatrix}
0 & 0 & 0 & M_{xxyz} & 0 & 0 & M_{xxzy} & 0 & 0 \\
0 & 0 & 0 & M_{yyyz} & 0 & 0 & M_{yyzy} & 0 & 0 \\
0 & 0 & 0 & M_{zzyz} & 0 & 0 & M_{zzzy} & 0 & 0 \\
M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & M_{zxxy} & 0 & 0 & M_{zxyx} \\
0 & 0 & 0 & 0 & M_{xyzx} & 0 & 0 & M_{xyxz} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{xxyz} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + \mathbb{M}_{yz}^{(3)} + \mathbb{T}_x^{(3)} + 2\mathbb{T}_x^{(4)} + 2\mathbb{T}_x^{\alpha(2)} - 4\mathbb{T}_x^{\beta(1)} \\
M_{xxzy} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + \mathbb{M}_{yz}^{(3)} - \mathbb{T}_x^{(3)} - 2\mathbb{T}_x^{(4)} - 2\mathbb{T}_x^{\alpha(2)} - 4\mathbb{T}_x^{\beta(1)} \\
M_{yyyz} &= \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)} + 2\mathbb{T}_x^{(1)} + \mathbb{T}_x^{(3)} - 2\mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\alpha(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_x^{\beta(2)} \\
M_{yyzy} &= \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} + 2\mathbb{M}_{yz}^{(5)} + 2\mathbb{T}_x^{(1)} - \mathbb{T}_x^{(3)} - 2\mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\alpha(2)} + 2\mathbb{T}_x^{\beta(1)} - \mathbb{T}_x^{\beta(2)} \\
M_{zzyz} &= -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} + 2\mathbb{M}_{yz}^{(5)} - 2\mathbb{T}_x^{(1)} + \mathbb{T}_x^{(3)} + 2\mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\alpha(2)} + 2\mathbb{T}_x^{\beta(1)} - \mathbb{T}_x^{\beta(2)} \\
M_{zzzy} &= -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)} - 2\mathbb{T}_x^{(1)} - \mathbb{T}_x^{(3)} + 2\mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\alpha(2)} + 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_x^{\beta(2)} \\
M_{yzxx} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} - \mathbb{M}_{yz}^{(3)} + 4\mathbb{T}_x^{\beta(1)} \\
M_{yzyy} &= \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(3)} - 2\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)} - 2\mathbb{T}_x^{\beta(1)} \\
M_{yzzz} &= -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(3)} + 2\mathbb{T}_x^{(1)} - 2\mathbb{T}_x^{\alpha(1)} - 2\mathbb{T}_x^{\beta(1)} \\
M_{zxxy} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(5)} - \mathbb{T}_x^{(1)} + \mathbb{T}_x^{(4)} - 4\mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\alpha(2)} - \mathbb{T}_x^{\beta(2)} \\
M_{zxyx} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(5)} - \mathbb{T}_x^{(1)} - \mathbb{T}_x^{(4)} - 4\mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\alpha(2)} + \mathbb{T}_x^{\beta(2)} \\
M_{xyzx} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(5)} + \mathbb{T}_x^{(1)} + \mathbb{T}_x^{(4)} + 4\mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\alpha(2)} + \mathbb{T}_x^{\beta(2)} \\
M_{xyxz} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(5)} + \mathbb{T}_x^{(1)} - \mathbb{T}_x^{(4)} + 4\mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\alpha(2)} - \mathbb{T}_x^{\beta(2)}
\end{aligned}
$$

* Rank 4 tensor (a).

$$
\begin{bmatrix}
M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & M_{zxxy} & 0 & 0 & M_{zxyx} \\
0 & 0 & 0 & 0 & M_{xyzx} & 0 & 0 & M_{xyxz} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{yzxx} &= \mathbb{T}_x^{(5)} + 2\mathbb{T}_x^{(6)} + 2\mathbb{T}_x^{\alpha(3)} \\
M_{yzyy} &= -2\mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(5)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{yzzz} &= 2\mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(5)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)} \\
M_{zxxy} &= 2\mathbb{M}_{yz}^{(4)} + \mathbb{M}_{yz}^{(6)} - \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{zxyx} &= -2\mathbb{M}_{yz}^{(4)} + \mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} + \mathbb{T}_x^{\beta(3)} \\
M_{xyzx} &= 2\mathbb{M}_{yz}^{(4)} - \mathbb{M}_{yz}^{(6)} + \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)} \\
M_{xyxz} &= -2\mathbb{M}_{yz}^{(4)} - \mathbb{M}_{yz}^{(6)} - \mathbb{T}_x^{(2)} + \mathbb{T}_x^{(6)} - \mathbb{T}_x^{\alpha(3)} - \mathbb{T}_x^{\beta(3)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
0 & 0 & 0 \\
0 & 0 & M_{yyyz} \\
0 & M_{zzzy} & 0 \\
0 & M_{yyzy} & 0 \\
0 & 0 & 0 \\
0 & 0 & M_{xxyz} \\
0 & 0 & M_{yzzz} \\
0 & M_{zxxy} & 0 \\
0 & 0 & 0 \\
-\frac{M_{yyyz}}{2} + \frac{M_{yyzy}}{2} + M_{zxxy} & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{yyyz} &= \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)} \\
M_{zzzy} &= -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} - 2\mathbb{M}_{yz}^{(5)} \\
M_{yyzy} &= \mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} + \mathbb{M}_{yz}^{(3)} + 2\mathbb{M}_{yz}^{(5)} \\
M_{xxyz} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + \mathbb{M}_{yz}^{(3)} \\
M_{yzzz} &= -\mathbb{M}_{4x}^{\alpha(1)} - \mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(1)} + 2\mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(3)} \\
M_{zxxy} &= 2\mathbb{M}_{4x}^{\beta(1)} + \mathbb{M}_{yz}^{(2)} - \mathbb{M}_{yz}^{(5)}
\end{aligned}
$$